

Design Task - Lorry Crane Hydraulics

REPORT FOR DESIGN TASK IN FLUID POWER SYSTEMS

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Abstract: The following report consists of the design and mathematical process which we underwent to design the hydraulic circuit for a lorry crane. The system contains two cylinder, and it was our task to determine the operating conditions for the cylinders and pump, and pick-out the appropriate components for the task. The dimensioning also required us to consider a work-case scenario, which informed our decision during part-selection.

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1. Design Task Definition

The objective of the design is to design a lorry crane, that can operative with a given mass over a defined drive cycle, and withstand a worst-case loading scenario.

For the task a certain values were given, mainly defining the drive cycle. Some values were true for all group, while others were group specific.

1.1. General Values

The general values for the drive-cycle provided included:

- $v_{p01} \rightarrow$ Normalised velocity vector for piston 1
- $v_{p02} \rightarrow$ Normalised velocity vector for piston 2
- $e_1 \rightarrow$ Level-arm for cylinder 1
- $e_2 \rightarrow$ Level-arm for cylinder 2
- $\theta_1 \rightarrow$ First arm angle
- $\theta_2 \rightarrow$ Second arm angle
- $L_0 \rightarrow$ Base height for lorry crane
- $p_2 \rightarrow$ Pressure on low-pressure side
- $\eta \rightarrow$ Oil dynamic viscosity
- $c_v \rightarrow$ Laminar leakage loss coefficient
- $k_p \rightarrow$ Coloumb friction coefficient
- $k_v \rightarrow$ Viscous friction loss coefficient
- $k_\epsilon \rightarrow$ Displacement coefficient

The information on the work-case scenario, provided the following values:

Parameter	Cylinder 1	Cylinder 2
$\theta_{1,wc}[\text{rad}]$	1,2	0,35
$\theta_{2,wc}[\text{rad}]$	0,0	0,0
$e_{1,wc}[\text{m}]$	0,14	0,48
$e_{2,wc}[\text{m}]$	0.57	0,17

1.2. Group-specific values

There were many group-specific values that we were given for the design task, these included:

- $M_{t,wc} \rightarrow$ Maximum total load mass
- $q_{r,wc} \rightarrow$ Worst-case flow to drive-cycle-max flow ratio
- $M_t \rightarrow$ Total load mass
- $L_1 \rightarrow$ First arm length
- $L_2 \rightarrow$ Second arm length
- $v_{p0} \rightarrow$ Reference cylinder velocity
- $t_0 \rightarrow$ Drive-cycle Time
- $P_{s,max} \rightarrow$ Maximum system pressure
- $\eta_c \rightarrow$ Cylinder Efficiency

1.3. Denormalisation and discretization

From the provided values we can see that the velocity profile which we were provided has been normalised and we have been provided a reference value. Using this reference value we denormalised the normalised velocity profile, to determine the effective velocity profile of the two cylinders.

$$v_{p1} = v_{p0} \cdot v_{p01}$$

$$v_{p2} = v_{p1} \cdot v_{p02}$$

Additionally, we discretized time to a number of discrete times equal to the length of the provided vectors for velocity, angles and arms, as such:

```
time = linspace(0,t_0,length(e_1));
```

2. Cylinder Sizing and Selection

2.1. Crane Kinematics

Before finding the forces on the cylinder we have to determine the kinematics of the crane tip throughout the drive-cycle. We found the x and z position of the crane tip with the following equations:

$$x = L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_2)$$
$$z = L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_2) + L_0$$

Using these two values we can create two graphs, one of the tip height through time and one of the position of the crane in space through the cycle.

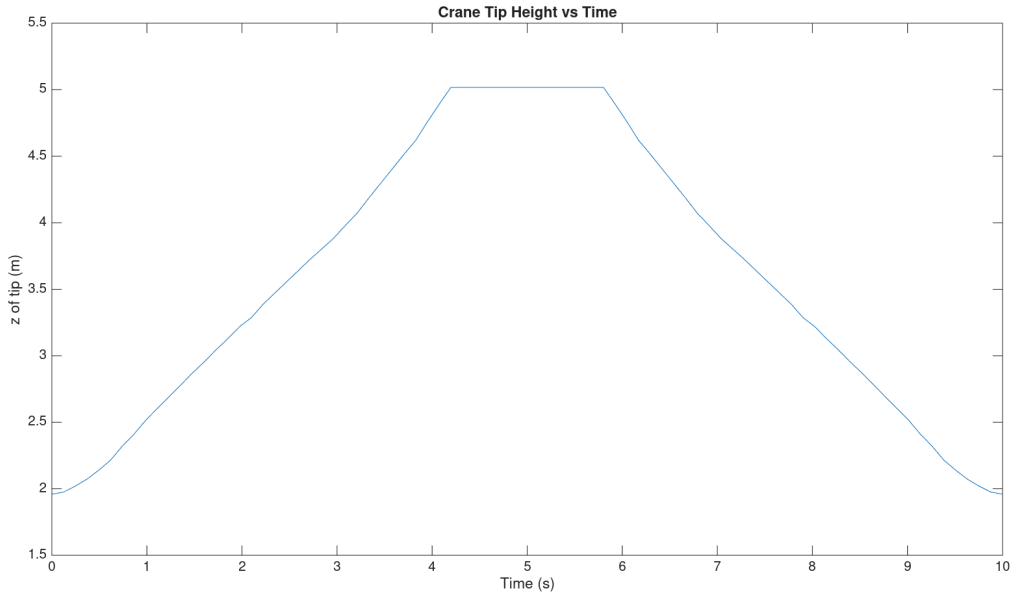


Figure 1: Height of crane tip with respect to time

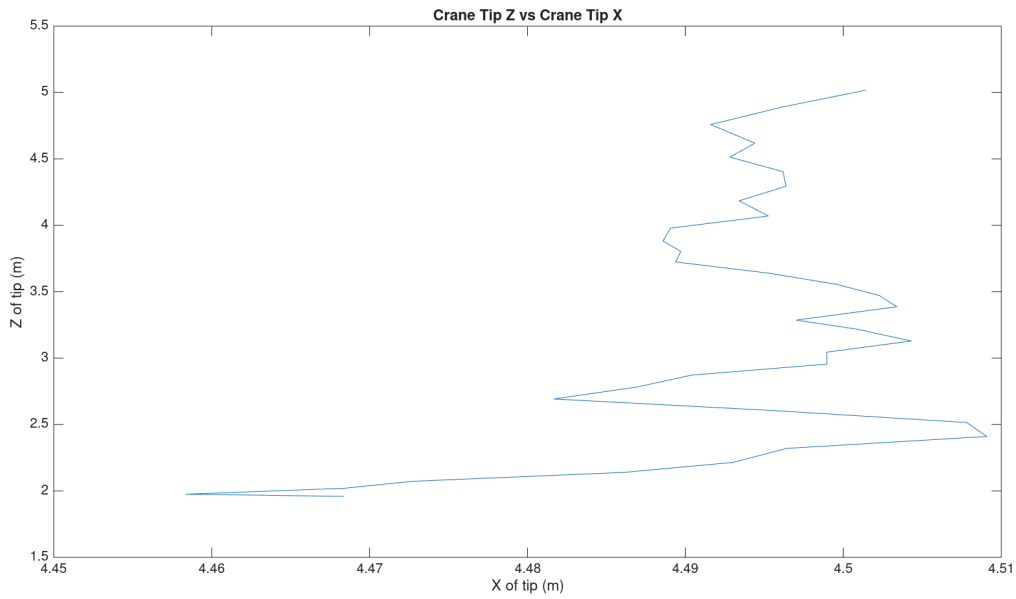


Figure 2: Crane tip position in space during drive cycle

2.2. Cylinder Drive-cycle

From here we continued by determining the forces on the two cylinders throughout the drive-cycle. We did this by calculating the moment generated on either arm hinge by the respective cylinder and the mass at the crane tip. We determined the resulting forces to have the equations:

$$F_1 = M_t \cdot g \cdot \frac{(L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_2))}{e_1}$$
$$F_2 = M_t \cdot g \cdot \frac{L_2 \cdot \cos(\theta_2)}{e_2}$$

The equivalent MATLAB code is as follows:

```
F_1 = M_t*g*(L_1*cos(th_1)+L_2*cos(th_2))./e_1;  
F_2 = M_t*g*L_2*cos(th_2)./e_2;
```

With F_1 and F_2 determined, we could now plot how they changed through time.

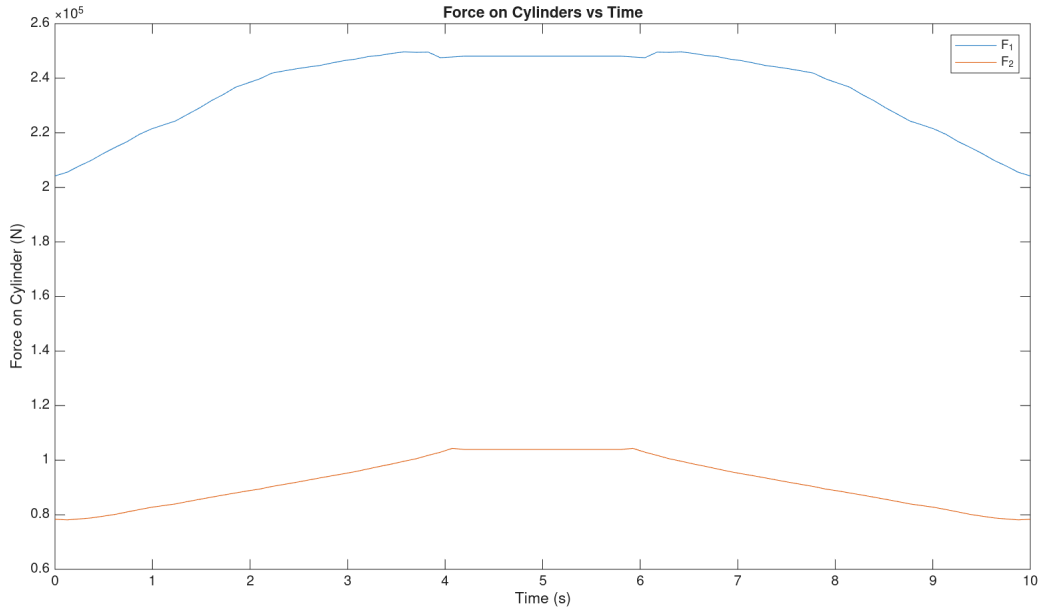


Figure 3: Forces on the two cylinders in the system

2.3. Worst-case Scenario and Cylinder Diameter

Using the same equations we used to determine the forces during the drive-cycle, we can find the forces in the worst-case scenario for the system, with the MATLAB code as follows:

```
F_wc_1 = M_tmax*g*(L_1*cos(th_1_wc_cyl1)+L_2*cos(th_2_wc_cyl1))./e_1_wc_cyl1;  
F_wc_2 = M_tmax*g*L_2*cos(th_2_wc_cyl2)./e_2_wc_cyl2;
```

In this case the θ and e values for either cylinder during the drive-cycle were replaced by the given worst-case scenario values. From these forces we could determine the diameter of the cylinder required to withstand these inopportune forces. We calculated the area and diameter required as:

$$A_1 = \frac{F_{1,wc}}{\eta_c \cdot (p_{1,max} - 0.5p_2)}$$

$$D_1 = \sqrt{\frac{4 \cdot A_1}{\pi}}$$

$$A_2 = \frac{F_{2,wc}}{\eta_c \cdot (p_{1,max} - 0.5p_2)}$$

$$D_2 = \sqrt{\frac{4 \cdot A_2}{\pi}}$$

In which $p_{1,max} = p_{s,max} - p_{extra}$, where p_{extra} is the unavoidable losses in the system, which we have to account for and which we arbitrarily determined to be $p_{extra} \approx 2000000 Pa$. We determined the following values for the respective variables:

- $A_1 = 0.020437 \text{ m}^2$
- $D_1 = 0.1613 \text{ m}$
- $A_2 = 0.014064 \text{ m}^2$
- $D_2 = 0.1338 \text{ m}$

Because no manufacturer produces cylinder of every size, we need to determine the cylinder with diameter which fits our needs. To do this we looked through *Bosch Rexroth's* linear hydraulic cylinder catalogue, and found a suitable candidate. Out of the catalogue we ended up choosing a two Series CGH3 type hydraulic cylinders. These cylinders are mill-type, double acting and can withstand the requirement maximum pressure. Of the CGH3 cylinder options we chose a cylinder with a diameter of 180mm and for the second cylinder a diameter of 140mm. In total the following data was determined necessary:

- Cylinder 1 Bore Diameter: $D_1 = 0.18 \text{ m}$
- Cylinder 2 Bore Diameter: $D_2 = 0.14 \text{ m}$
- Cylinder 1 Area on Piston Side: $A_{p1} = 0.025447 \text{ m}^2$
- Cylinder 1 Ring Area on Rod Side: $A_{r1} = 0.013175 \text{ m}^2$
- Cylinder 2 Area on Piston Side: $A_{p2} = 0.015394 \text{ m}^2$
- Cylinder 2 Ring Area on Rod Side: $A_{r2} = 0.007540 \text{ m}^2$

To be absolutely sure we made the right choice we also found the maximum stroke velocities for the respective cylinders and determined them to be:

- $v_{1,max} = 0.14m/s$
- $v_{2,max} = 0.16m/s$

We then plotted these two values alongside the absolute value of the velocities of the cylinders during their drive-cycles.

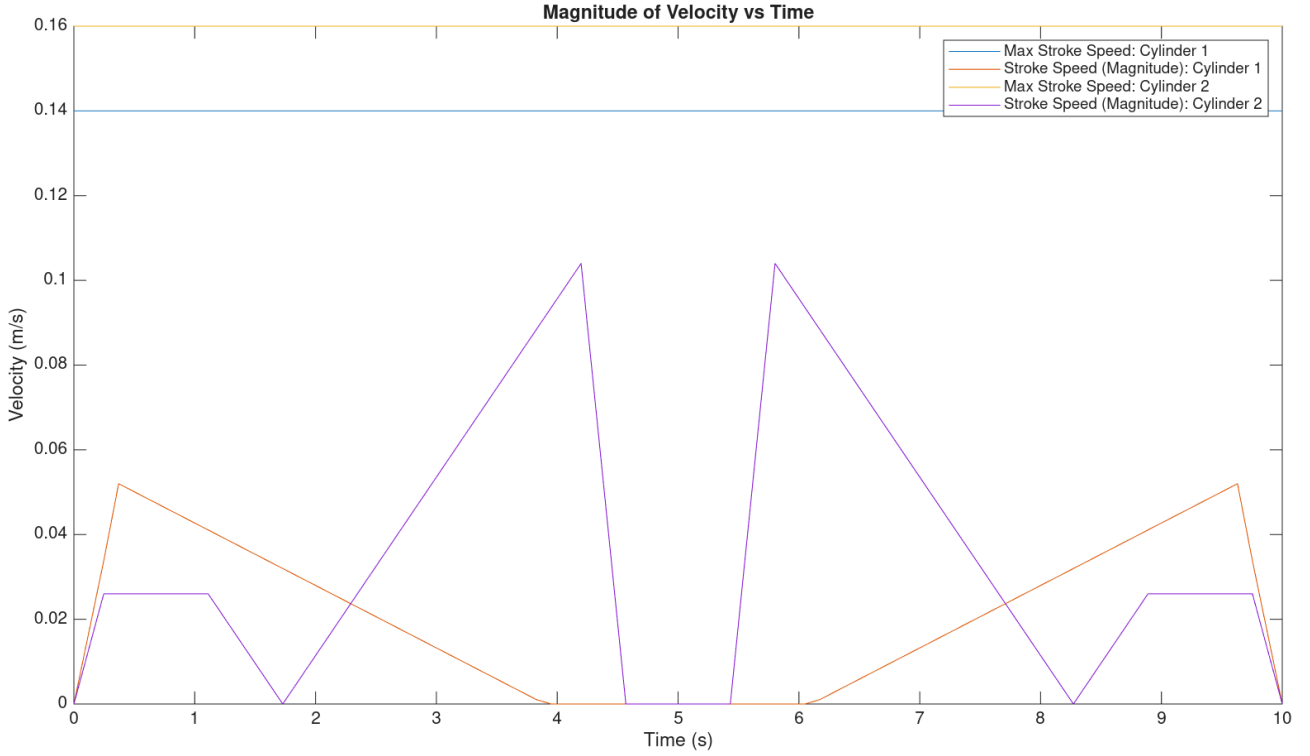


Figure 4: Determination of being under the maximum stroke velocity throughout cycle

We can see that we are under the maximum stroke velocities throughout the drive-cycle, so there is no issue with our choice.

3. Circuit Drive-cycle Analysis

3.1. System Pressure

For the drive-cycle analysis we need to determine the pressures and flow our cylinders operate at; this later help us determine what type of system hydraulic circuit we are interested in using. We know that the pressure on the low-pressure side will always be $1MPa$, and since we know that the sign of the force never changes on either cylinder, we know that the rod side of the cylinder will always be low-pressure side. Because of this we use one single value for p_2 for both cylinders. The equation to

determine the pressure on the high pressure side of the cylinder depends on whether the cylinder is acting as a pump or a motor. When the cylinder is operating as a pump the equation is:

$$p_1 = \frac{p_2 \cdot A_2 + \frac{F}{\eta_c}}{A_1}$$

And when the cylinder is operating as a motor, the equation is:

$$p_1 = \frac{p_2 \cdot A_2 + \eta_c \cdot F}{A_1}$$

Since the only difference is how the efficiency changes the resulting force, in MATLAB we can treat this equation as single equation, where η_c changes definition based on the sign of the velocity.

```

v_1_pos = v_p1 > 0;
v_1_neg = v_p1 < 0;
v_1_stat = v_p1 == 0;
eta_c1 = v_1_pos./eta_c + v_1_neg*eta_c + v_1_stat;
v_2_pos = v_p2 > 0;
v_2_neg = v_p2 < 0;
v_2_stat = v_p2 == 0;
eta_c2 = v_2_pos./eta_c + v_2_neg*eta_c + v_2_stat;

p_2 = p_2*ones(1,length(time));
p_1_cyl1 = (F_1./eta_c1 + p_2*A_r1)/(A_p1);
p_1_cyl2 = (p_2*A_r2 + F_2./eta_c2)/(A_p2);

```

We can also determine the minimum pressure our a constant-pressure pump would have to operate at for the system to function as:

```

p_1max_cyl1 = max(p_1_cyl1);
p_1max_cyl2 = max(p_1_cyl2);
p_min = max([p_1max_cyl1 p_1max_cyl2]) * ones(1,length(time));

```

From these computations we can plot the graph of pressure for the cycle and minimum pressure for the system.

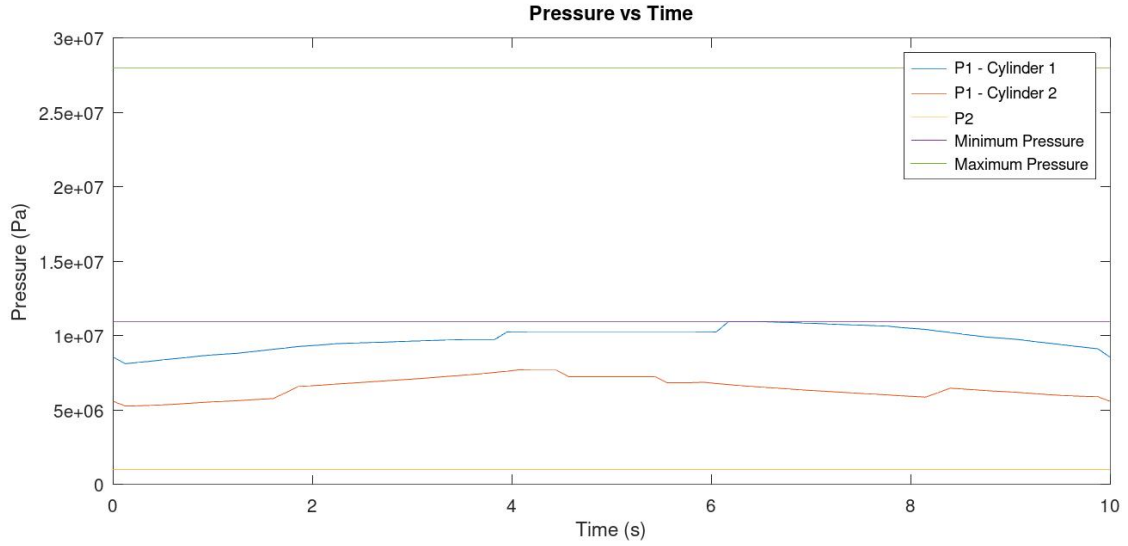


Figure 5: Pressure in 4 chambers, maximum system pressure, and minimum constant system pressure

3.2. System Flow

To determine the flow we also need to consider the sign which the velocity takes during the drive-cycle, as we have to change which chamber the flow moves to. The resulting MATLAB code is as follows.

```

q_1 = A_p1*v_p1.*v_1_pos + A_r1*abs(v_p1).*v_1_neg;
q_2 = A_p2*v_p2.*v_2_pos + A_r1*abs(v_p2).*v_2_neg;
q_tot = q_1 + q_2;
q_max = q_tot/q_rwc;

```

The equation for q_{max} came from the definition for the worst-case scenario flow ratio, and is the worst-case level of flow that our pump must be able to handle.

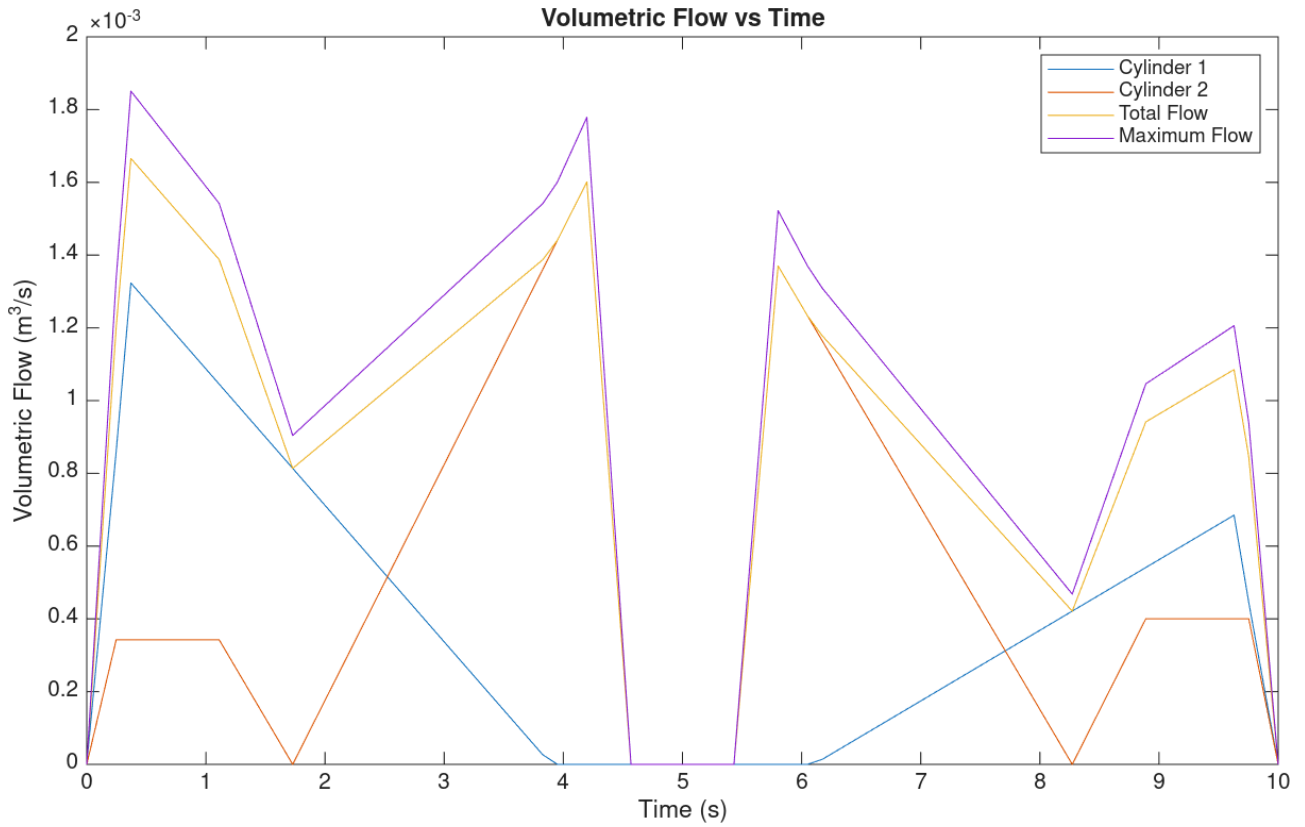


Figure 6: Change in Flow with Time

3.3. Hydraulic System Type and System Diagram

After our analysis of the system pressure and flow, we had to decide what type of circuit our hydraulic system would have. We decided that our hydraulic circuit should be a constant pressure system with variable flow. We made this choice because as can be seen in the respective plots, the pressure is relatively constant when compared to the flow which goes from $0.0016\text{m}^3/\text{s}$ to zero. Since all of the flow during the period where the velocity is zero would be wasted, we believed that the best way to reduce our losses would be to employ this system. We also considered that a more efficient system to reduce the losses further would be a load-sensing system with variable flow. But considering these types of system are even more expensive than the constant-pressure system, we decided to opt for a constant-pressure system to keep that system cost to a more middling cost. Finally the drivability of the system is relatively good, because it exhibit load dependency, which adds high damping to our system making it more drivable. The system also doesn't exhibit load interference which would reduce the drivability.

After choosing the type of system that we wanted we drew a possible circuit diagram for it.

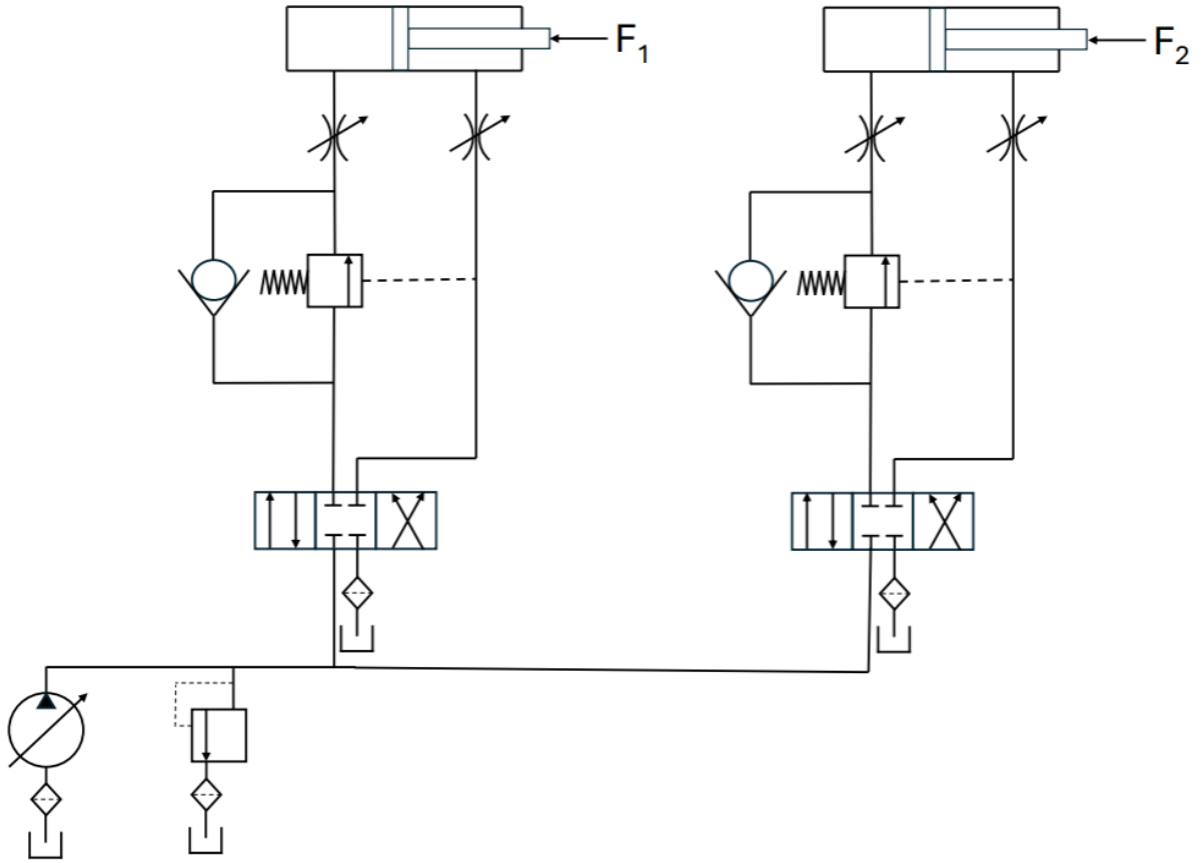


Figure 7: Possible Circuit Diagram

4. Pump Selection

For pump selection we decided to turn once again to the catalogue of Bosch Rexroth. Looking through their catalogue we found a pump that seemed to fit our requirement. This pump is a A10VZO size 45 pump. This pump has the following parameters:

- Maximum Displacement per Revolution, $V_{g,max} = 45 \text{ cm}^3/\text{rev}$
- Maximum Flow (at n_{nom} and $V_{g,max}$) $q_v = 135 \text{ [lpm]}$
- Maximum Rotational Speed, $n_{nom} = 3000 \text{ rpm}$
- Maximum Pressure, $P_{max} = 350 \text{ bar}$
- Nominal Pressure, $P_{nom} = 315 \text{ bar}$

The pump can operate up to it's maximum rotational speed in either direction. But when the rotational velocity is positive (when it's operating as a pump), and under 200 rpm, the operating period can only be below 3 minutes, if the displacement is $V_g \leq V_{g,max}$. The volumetric flow can be calculated with the

following equation:

$$q_v = \frac{V_{g,max} \cdot n \cdot \eta_{vol}}{1000} \text{ [lpm]}$$

5. System Analysis and Efficiency

The pump maintains a constant system pressure of 13 MPa, which is just above the maximum pressure required throughout the drive-cycle to allow some protection from pressure fluctuations. The graph as expected will just be a constant value:

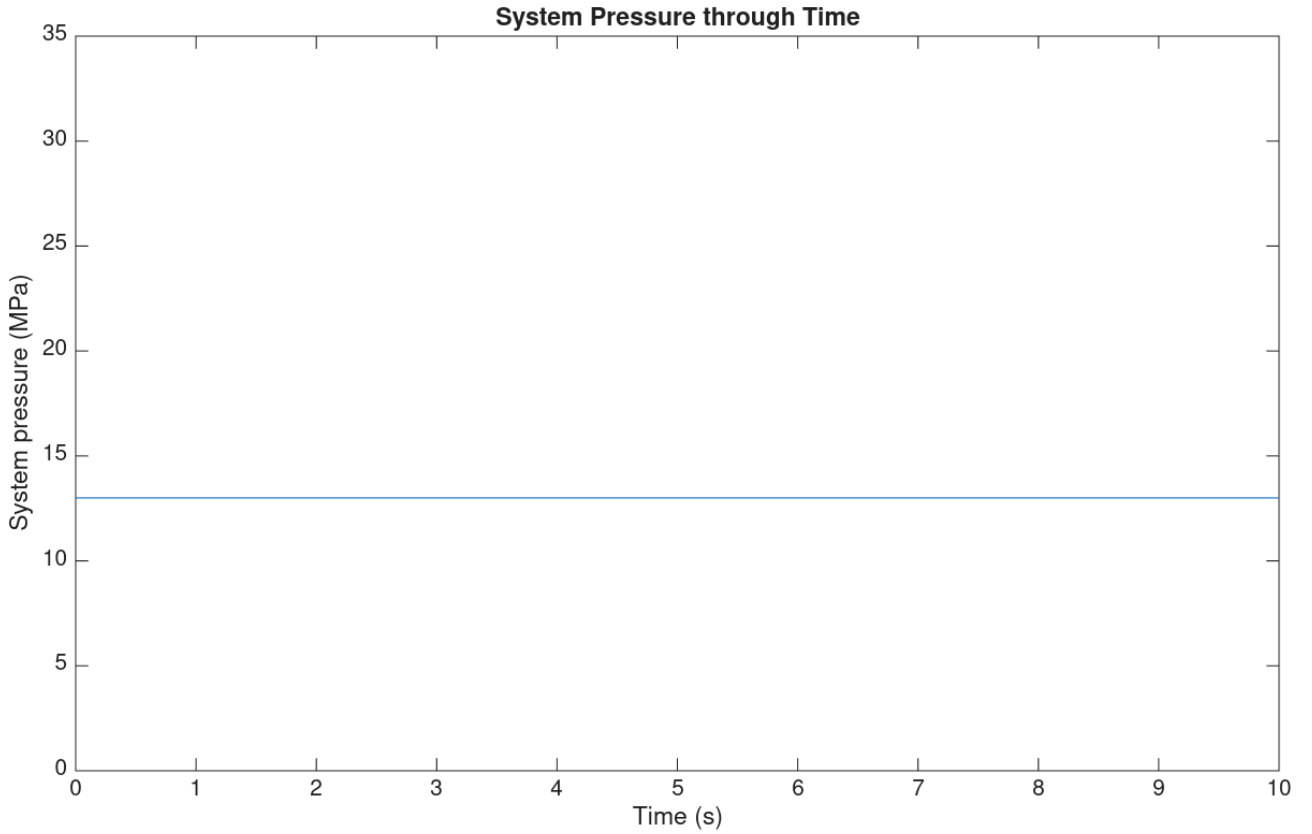


Figure 8: System pressure provided by pump

As we've chosen the variable flow pump, the flow will follow the maximum flow required by the system, as follows:

Assuming we are operating at a constant $n = n_{nom}$, we can derive the displacement for the pump as:

$$V_g = \frac{1000 \cdot q_v}{n_{nom} \cdot \eta_v}$$

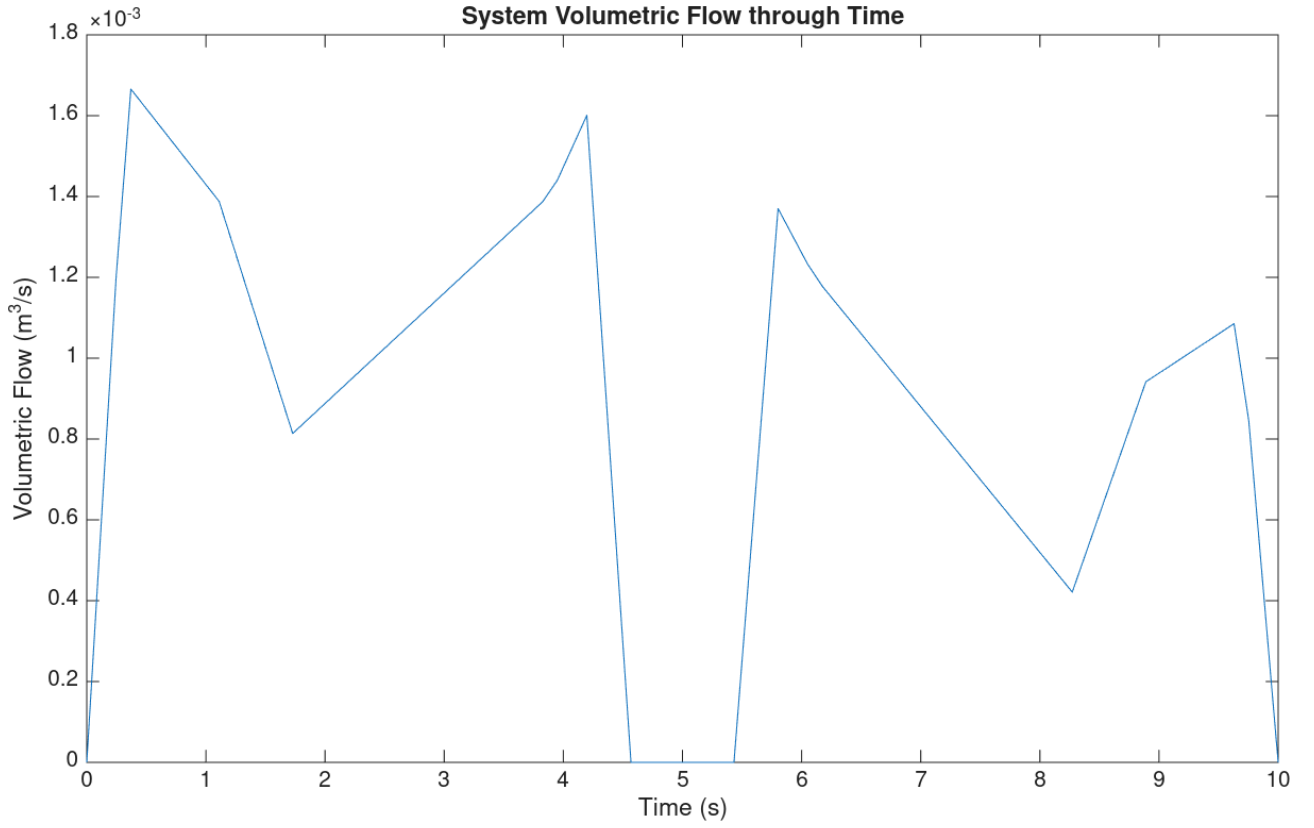


Figure 9: System Flow provided by pump

5.1. Displacement Coefficient and Efficiencies

To determine the volumetric and hydro-mechanical efficiency we need to find the displacement coefficient for the pump. To determine this coefficient we used the flow equation from the formula book.

$$q_{ep} = \varepsilon \cdot D_p \cdot n \cdot \eta_{vol,p}$$

The problem now was that we did not have some value, so we relied on some assumptions to determine these values. The first assumption is that $D_p = V_{g,max}$, additionally we also assumed that $n = 2500$ [rpm], and finally we assumed that $q_{ep} = q_{tot}$. Since we are determining the volumetric efficiency from this value, we couldn't assume a value, so what we decided to do was to plug in the equation for the efficiency itself:

$$\eta_{vol,p} = 1 - C_v \cdot \frac{\Delta p}{|\varepsilon_p| \cdot n_p \cdot \eta}$$

Plugging in this equation and isolating ε_p , we got the following equation:

$$\varepsilon_p = \frac{q_{tot} + V_{g,max} \cdot C_v \cdot P_{nom} / \eta}{V_{g,max} \cdot n}$$

For the volumetric and hydro-mechanical we used the equations as indicated in the formula book,

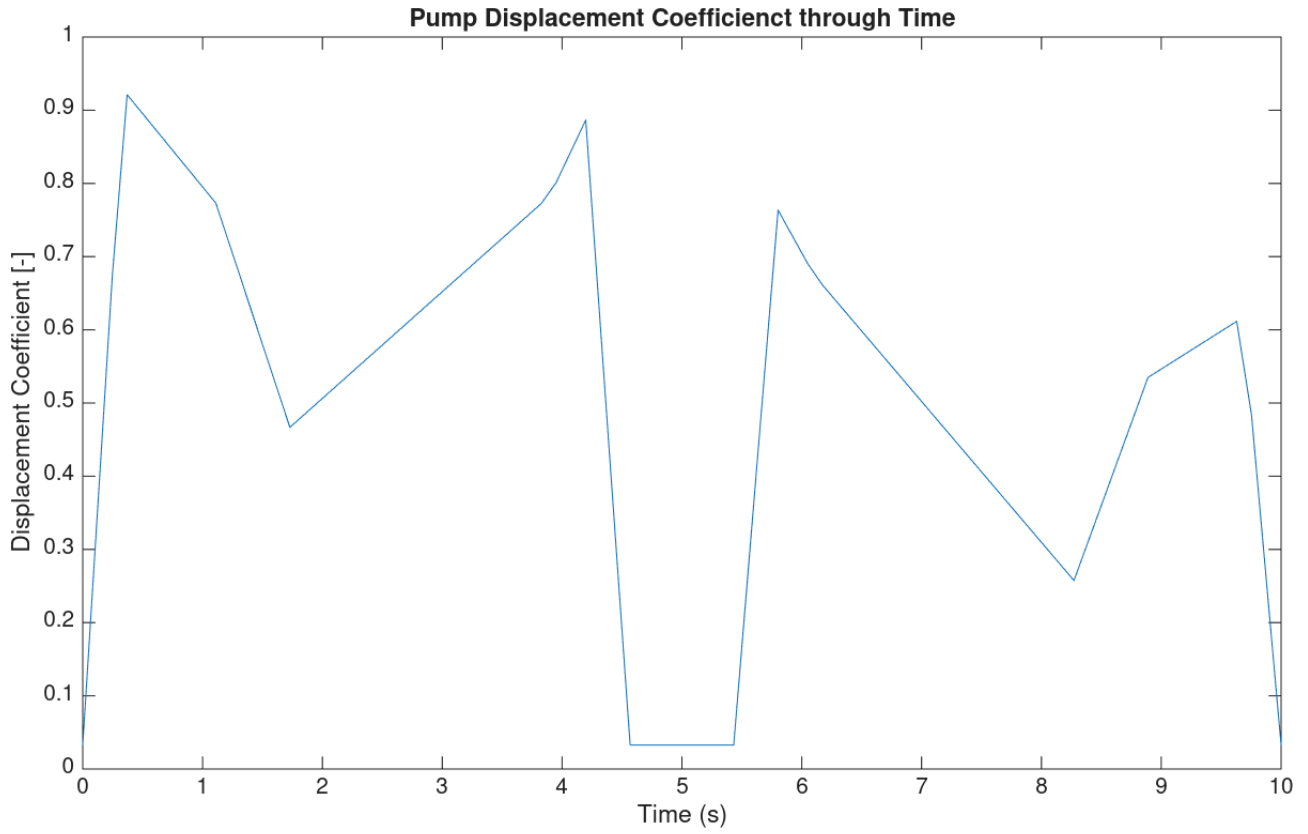


Figure 10: Displacment Coefficient

which we will report.

$$\eta_{vol,p} = 1 - C_v \cdot \frac{\Delta p}{|\varepsilon_p| \cdot n_p \eta}$$

$$\eta_{hm,p} = \frac{1}{1 + (k_p \cdot k_v \frac{n_p \eta}{\Delta p}) \cdot e^{(k_\varepsilon(1-|\varepsilon_p|))}}$$

And the total efficiency is obviously product of the two efficiencies:

$$\eta_{tot} = \eta_{vol} \cdot \eta_{hm}$$

From the values we found in the section we were able to determine the hydraulic input power:

$$P_{in} = q_{tot} \cdot \Delta p_{nom} ./ \eta_{tot} - v_{1_neg} .* F_{1_1} \cdot v_{p1} - v_{2_neg} .* F_{2_1} \cdot v_{p2} ;$$

We also determined the output power with the following MATLAB code:

$$P_{out} = v_{1_pos} \cdot v_{p1} \cdot F_{1_1} + v_{2_pos} \cdot v_{p2} \cdot F_{2_1} ;$$

We determined that while the cylinder acted as a pump, the mechanical output power was equal to $v \cdot F$, but when the cylinder acted as a motor the power can be interpreted as an input power because it is acting onto the system rather than extracting power from the system. The power graph became the following:

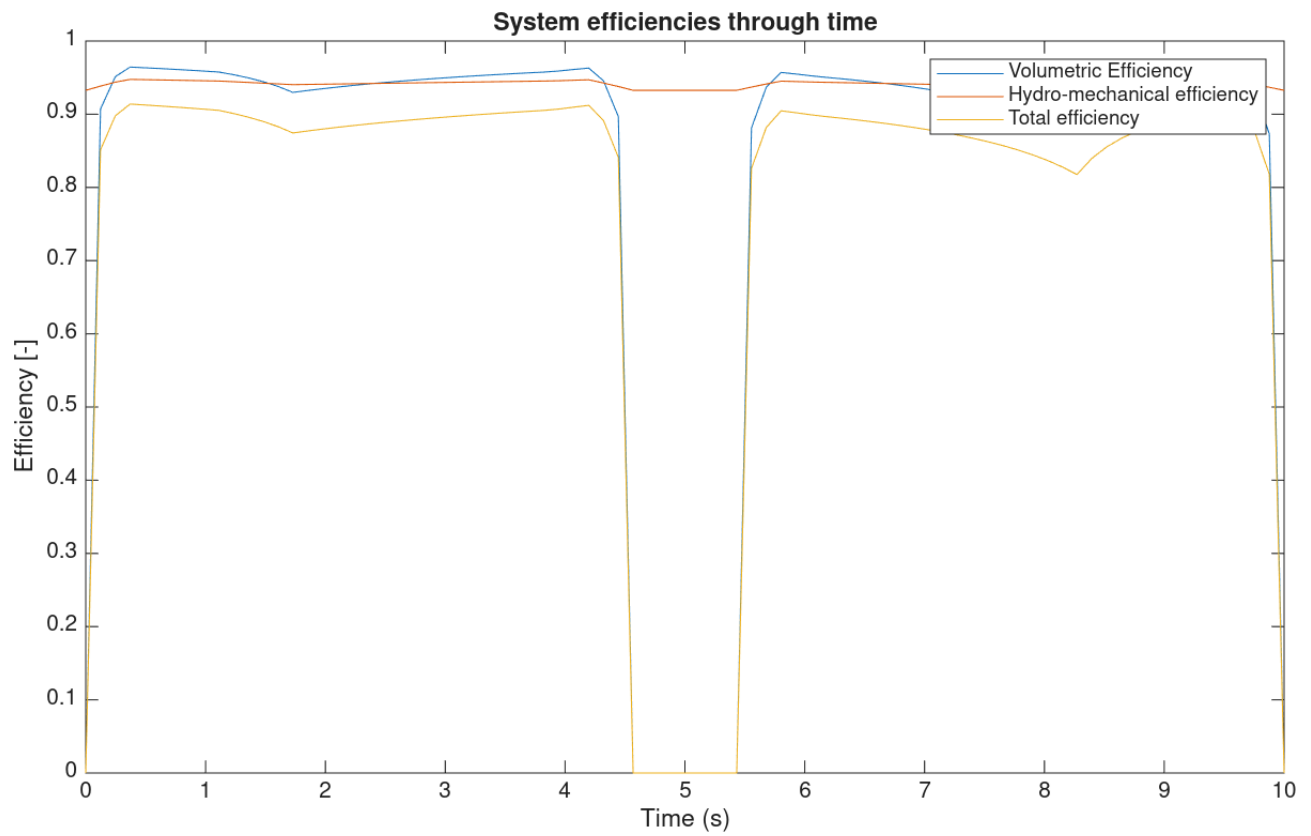


Figure 11: Volumetric, Hydro-mechanical and Total Efficiency

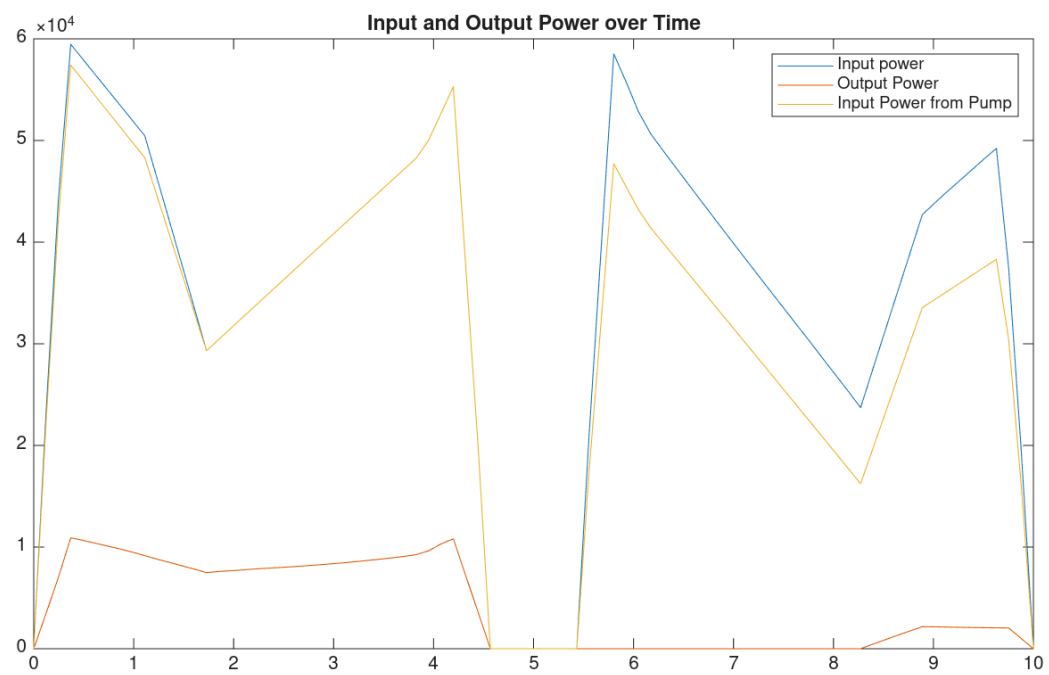


Figure 12: Input and Output Power for System

With the resulting efficiency being determined by the fraction of input power that becomes output power.

$$\eta_{\text{sys}} = \frac{P_{\text{out}}}{P_{\text{in}}}$$

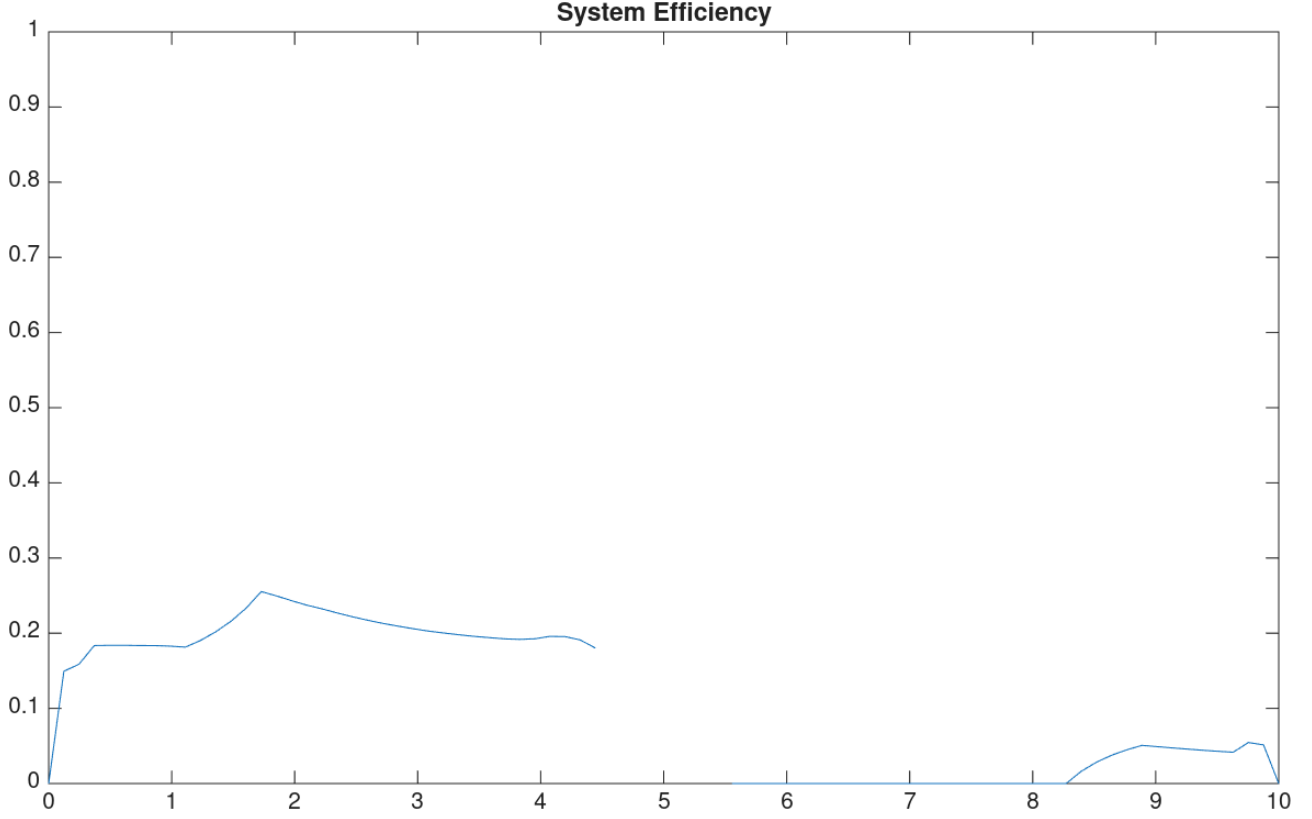


Figure 13: Input and Output Power for System

The efficiency of the system is simply abysmal, but that is to be expected from a constant pressure system. We chose the system for it's relative design simplifity and as a middle-ground between the load-sensing system and constant flow system. The efficiency after around 6 seconds is zero, not because the system is not in motion, but because cylinders are both acting as motors, and so the mechanical power can be considered a power input rather than output, resulting in the output power being zero. A way the energy loss could be reduced slightly is to change the system such that the rod side of the cylinders suck fluid from the tank rather than being filled by the pump, this would allow the displacement coefficient to be zero and reduce some of the losses in the system.

The efficiency is relatively when compared to other systems such as a load-sensing system. If we had used a load-sensing system we would have higher efficiency, and probably better drivability, but the cost would also greater. Whereas in term of efficiency the constant flow system would probably have lower efficiency in the first-half, but higher efficiency on the second-half. We would not recommend our system because of the very low efficiencies on the return-stroke that also require more expenses on the cooler.

6. Cooler Dimensioning

The overall work performed by the machine is 0 because the overall displacement of the crane comes out to be 0 at the end of the drive-cycle. Because the effective work is 0, this means that all the power that placed into the system is lost. This means that all the input power eventually becomes heat, and so the average cooling power that we need to supply is the average of the input power.

$$P_{avg} = mean(P_{in})$$

The average power wasted is approximately $35783W$. So our active cooler has to have the ability to extract this much power to be able to keep the system at normal operating temperatures.